

Task 1: Creating a React Application:

```
PS D:\React Working> npm create vite@latest myfirstreact
Ok to proceed? (y) y

> npx
> create-vite myfirstreact

|
| ◊ Select a framework:
|   React
|
| ◊ Select a variant:
|   JavaScript
|
| ◊ Which linter to use?
|   ESLint
|
| ◊ Install with npm and start now?
|   Yes
|
| ◊ Scaffolding project in D:\React Working\myfirstreact...
|
| ◊ Installing dependencies with npm...
|
added 135 packages, and audited 136 packages in 13s

31 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities

|
| ◊ Starting dev server...
|
> myfirstreact@0.0.0 dev
> vite

VITE v8.1.5 ready in 497 ms
→ Local: http://localhost:5173/
```

Why we are using the Vite.js?

We are using Vite because the current version of Node.js (v24.15.0) is relatively new, while Create React App (CRA) has been deprecated and is no longer actively maintained. As a result, CRA was not designed or tested for compatibility with the latest Node.js versions and may cause installation or performance issues. Vite is the modern, officially recommended tool for setting up React applications, offering faster project creation, better compatibility with newer Node.js releases, and improved development performance.

Code:

App.jsx

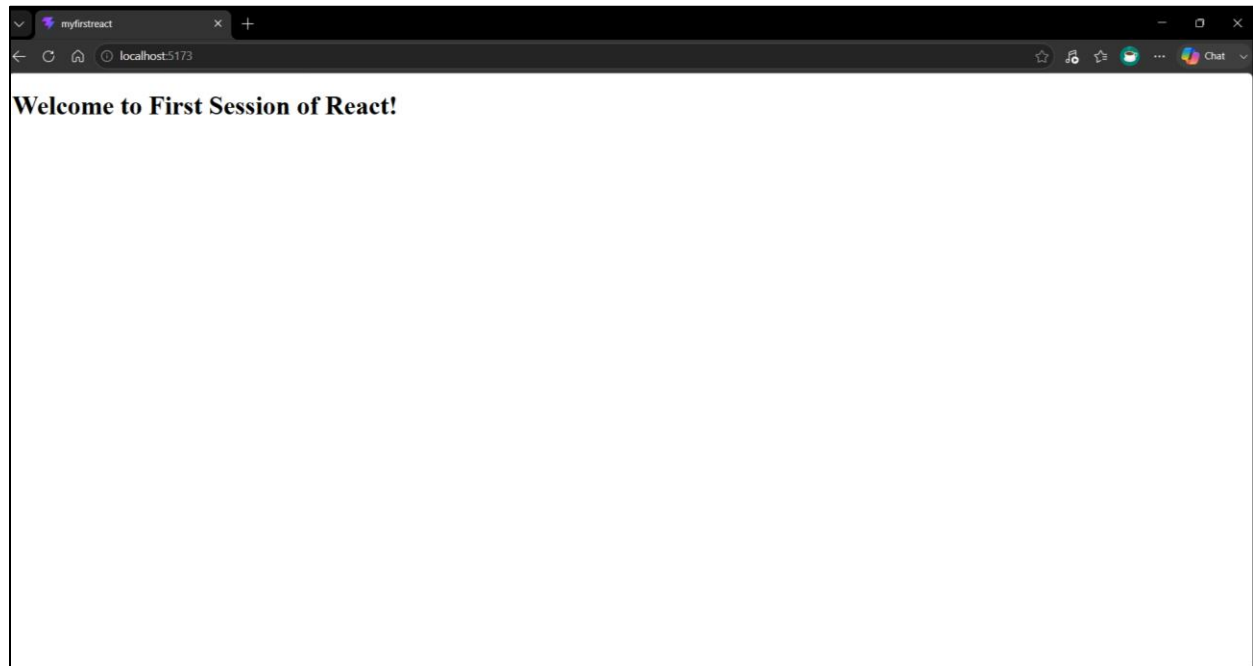
```
function App(){
  return(

    <h1>Welcome to First Session of React!</h1>

  )
}
```

```
export default App;
```

Output:



Theory Question Answers:

1. What is SPA (Single Page Application)?

A Single Page Application loads a single HTML page and dynamically updates the content without refreshing the entire page.

Benefits of SPA

- Faster page loading.
- Better user experience.
- Reduced server load.
- Smooth navigation between pages.
- Efficient use of APIs

What is React?

React is an open-source JavaScript library developed by Facebook for building interactive user interfaces, especially SPAs.

Working of React

- Creates reusable UI components.
- Uses Virtual DOM for efficient rendering.
- Updates only the changed parts of the webpage.
- Provides fast and responsive user interfaces.

SPA vs MPA

SPA	MPA
Loads one page	Loads multiple pages
Faster navigation	Slower navigation
Better user experience	Traditional navigation
Uses APIs extensively	Server renders pages
Examples: Gmail, Netflix	Examples: Amazon (traditional), news websites

Virtual DOM

Virtual DOM is a lightweight copy of the real DOM maintained by React.

Advantages

- Faster rendering.
- Improves application performance.
- Updates only the modified elements.
- Reduces unnecessary DOM manipulations.

Features of React

- Component-based architecture.
- Virtual DOM.
- One-way data binding.

- Reusable components.
- High performance.
- Easy integration.
- Large community support.